

Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination**COMPUTER ARCHITECTURE**

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
- (2) Due credit will be given to neatness.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (a) A benchmark program is run on a 40 MHz processor. The executed program consists of 100,000 instruction executions, with the following instruction mix and clock cycle count :

Instruction type	Instruction count	Cycles per Instruction
Integer arithmetic	45000	1
Data transfer	32000	2
Floating point	15000	2
Control transfer	8000	2

Determine the effective CPI, MIPS rate, and execution time for this program. 5(CO3)

- (b) Implement a 4-bit full adder in verilog using behavioural model. 5(CO4)
2. (a) Multiply (23) and (19) using Booth's algorithm. Where each number is represented using 6 bits. 5(CO2)
- (b) Explain IEEE 745 single precision and double precision floating point format with examples. 5(CO2)

3. Design a hardwired control unit for two's complement multiplier using delay element method. 10(CO5)
4. (a) Design 32K × 8-bit memory using 16K × 1 static memory chip. 5(CO5)
(b) Explain Four way and Eight way set associated cache memory considering a memory of size 1 KB. 5(CO1)
5. Explain in detail the concept and architecture of a Direct Memory Access system. 10(CO1)
6. Discuss pipeline processors and explain its types and design. 10(CO2)

